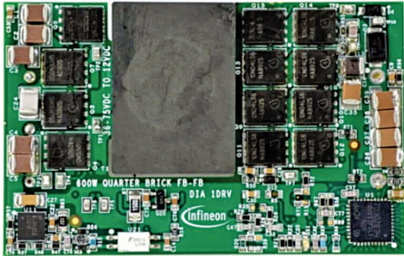


Nidhi Agarwal is Senior Technology Journalist at EFY with a deep interest in embedded systems, development boards, and IoT cloud solutions.

Full-Bridge 600W Isolated DC-DC



The power module gives 12V and 600W with high efficiency. It features smart control, easy setup, digital communication, and a small size for telecom and industrial use.

The REF_600W_FFB_XDPP1100 is an isolated DC-DC power module for telecom and industrial applications, delivering up to 96% efficiency in a DOSA-standard quarter-brick form factor. It uses a hard-switching full-bridge topology with secondary-side digital control, providing 12V at 600W output. The digital controller combines an analogue front end, pre-programmed peripherals, a fast control loop, and an embedded microcontroller to improve performance, reduce the bill of materials, and enable control features. The converter uses voltage mode control with built-in flux balancing and transformer flux walking protection, ensuring fast response to line and load changes for stable operation. Communication is via PMBus and I2C for digital system integration. Configurable through a graphical user interface, it allows firmware customisation, improving reliability and design flexibility. The integration of components enhances power density and reduces system costs. Pre-programmed control features enable fast time-to-market, while PMBus support allows adaptability to future needs.

The reference design is suitable for isolated DC-DC brick modules, server motherboards, 48V computing, and data centre power supplies.

Infineon

<https://www.electronicshobby.com/electronics-projects/full-bridge-600-w-isolated-dc-dc-reference-design>

Full-Bridge Converter



The reference design efficiently converts high-voltage power. It supports various voltage ranges, saves space, and is easy to adjust. Built-in control and safety features make it suitable for various power applications.

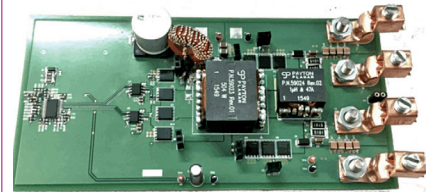
The STDES-6KWHVDCDC reference design from STMicroelectronics is a 6kW high-voltage DC-DC converter designed for EV charging infrastructure and other power applications in automotive, industrial, and renewable energy systems. It utilises a digital full-bridge LLC resonant converter architecture. It offers two hardware configurations: a dual-transformer setup for voltage ranges and a single-transformer version covering 200-1000V, both of which use the same PCB for easy modification. A frequency foldback control scheme enables efficiency under different load and voltage conditions. The design uses ST's silicon carbide diodes for output rectification and MDMesh DK5 power MOSFETs for high-voltage switching. Control is handled by the microcontroller, which operates the power stage through PM8834MTR gate driver ICs and gate drive transformers, supported by a secondary-referenced auxiliary power supply. Safety features and testing support the design's use as a tool for engineers working on high-voltage DC-DC power conversion systems.

The reference design is suitable for automotive, industrial, and renewable energy applications.

STMicroelectronics

<https://www.electronicshobby.com/electronics-projects/full-bridge-converter-reference-design>

Phase-Shifted Full-Bridge Converter



The reference design provides efficient 400W power conversion with zero-voltage switching. It offers fast load response and low standby power. Built-in isolation ensures safety.

The TIDA-01407 is a 400W reference design from Texas Instruments for automotive power systems, using a phase-shifted full-bridge converter with a 36-60V DC input and a 12V output. A controller with programmable delays enables zero-voltage switching across varying conditions. Synchronous rectifiers on the output enable fast load response and a wide loop bandwidth, maintaining a stable 12V output even at no load, with standby power under 100mW. A transformer provides isolation to prevent the 48V input from reaching the output in the event of a failure. Half-bridge gate drivers on the primary side control the MOSFETs and support a boot voltage of up to 120V. The output is regulated from the primary side, avoiding the need for optocouplers and reducing board space. The design includes three parts: primary-side current sensing using a transformer for overcurrent protection and delay control; the main power stage with switching and power modes based on load; and synchronous rectification on the secondary side which reduces voltage drops and manages load shifts. It supports power delivery in traction inverters, control units, and DC-DC converters in mild hybrid vehicles.

The reference design is suitable for automotive systems, such as traction inverters and converters.

Texas Instruments (TI)

<https://www.electronicshobby.com/electronics-projects/phase-shifted-full-bridge-converter-reference-design>